

A bag contains one rupee, 50 paise and 25 paise coins in the ratio 2 : 3 : 5. Their total value is Rs. 114. The value of 50-paise coins is:

How  
low.

A. Rs. 24

B. Rs. 36

C. Rs. 48

D. Rs. 72

E. None of these



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# TIME'S UP!

## Question :

A bag contains 50 P, 25 P and 10 P coins in the ratio 9: 4, amounting to Rs. 206. Find the number of coins of each type respectively.

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- A. 360, 160, 200
- B. 160, 360, 200
- C. 200, 360, 160
- D. 200, 160, 300

$$\begin{array}{r}
 50 \quad 25 \quad 10 \\
 5 \quad 9 \quad 4 \\
 \hline
 250 \quad 225 \quad 40 \\
 \hline
 \end{array}$$

$$1 - \frac{206 \times 100}{515} = \frac{20600}{515} = 40$$





In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there? **TCS 12<sup>th</sup> Sept**

- A. 50
- B. 100
- C. 150
- D. 200



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$$\begin{array}{r}
 10 \quad 5 \\
 2 \quad 3 \\
 \hline
 + 20 + 15 \\
 \hline
 \end{array}$$

$$1 - \frac{30}{60} \times 100 = 50$$

$$60p = 30 \times 100p$$



## Question: <sup>\* 16</sup>

A bag contains Rs. 4100 in the form of Rs. 5, Rs. 2 and Rs. 1 coin. The number of coins is in ratio 4: 6: 9. So, find the number of 1 Rs. coins.

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(a) 40

(b) 50

(c) 900

(d) 70

$$\begin{array}{r}
 5 \quad 2 \quad 1 \\
 4 \quad 6 \quad 9 \\
 \hline
 20 + 12 + 9
 \end{array}$$

$$\begin{array}{l}
 41 = 4100 \\
 \boxed{1 = 100} =
 \end{array}$$



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Question:.

A bag contains Rs. 4100 in the form of Rs. 5, Rs. 2 and Rs. 1 coin. The number of coins is in ratio 4: 6: 9. So, find the number of 1 Rs. coins.

(a) 40

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(c) 900

(d) 70

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Handwritten solution:

|     | 5  | 2  | 1 |
|-----|----|----|---|
| No. | 4  | 6  | 9 |
|     | 20 | 12 | 9 |

$41 = 4100$   
 $1 = 100$



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If Rs. 782 be divided into three parts, proportional to  $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$ , then the first part is:

A. Rs. 182

B. Rs. 190

C. Rs. 196

D. Rs. 204

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$$\begin{array}{ccc}
 12 & 18 & x = 27 \\
 & \uparrow \times 9 & \uparrow \times 9 \\
 & 2 & 3
 \end{array}$$

### Question

The third proportional of 12 and 18 is  
27

TCS 12th Sep

### Question

What is the third proportional to 10 and 20?

\_\_\_\_\_

### Question

The third proportional to 0.8 and 0.2 is :  
 \_\_\_\_\_



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$$\begin{array}{cc} 4 : 7 & 14 : 7 \\ \hline 0.12, 0.21, 8, x \end{array}$$

$$\begin{array}{l} 3 \sqrt{12} = 8^2 \\ 2 \sqrt{7} \quad x \\ \hline x = 14 \end{array}$$

## Question

The fourth proportional to 0.12, 0.21, 8 is:

TCS 2021 12th sept

## Question

What is the fourth proportional to 24, 120 and 22?

$$\begin{array}{ccc} 24, 120, 22, x & 110 \\ \hline 1:5 & 1:5 \end{array}$$

## Question

What is the fourth proportional to 189, 273 and 153?

$$\frac{24}{120} = \frac{22}{x}$$

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$$\begin{array}{cc} 4 : 7 & 14 : 7 \\ \hline 0.12 & 0.21 \end{array}$$

$$\frac{12}{21} = \frac{8}{x}$$

$$x = 14$$

### Question

The fourth proportional to 0.12, 0.21, 8 is:

TCS 2021 12th sept

### Question

What is the fourth proportional to 24, 120 and 22?

$$\frac{24}{120} = \frac{22}{x}$$

$$x = 110$$

### Question

What is the fourth proportional to 189, 273 and 153?

$$\frac{189}{273} = \frac{153}{x}$$

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$$\begin{array}{cc} 4 : 7 & 14 : 7 \\ \hline 0.12, 0.21, 8, x \end{array}$$

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TCS 2021 12th sept

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Raju and Lalitha originally had marbles in the ratio 4:9. Then Lalitha gave some of her marbles to Raju. As a result, the ratio of the number of marbles with Raju to that with Lalitha became 5:6. What fraction of her original number of marbles was given by Lalitha to Raju?

TCS Digital Type

CAT 2018[SLOT(1)]

R  
44  
55

L  
99  
78

13  
11

11  
13

a) 6/19

b) 7/33

c) 1/5

d) 1/4

$$\frac{21}{99} = \frac{7}{33}$$

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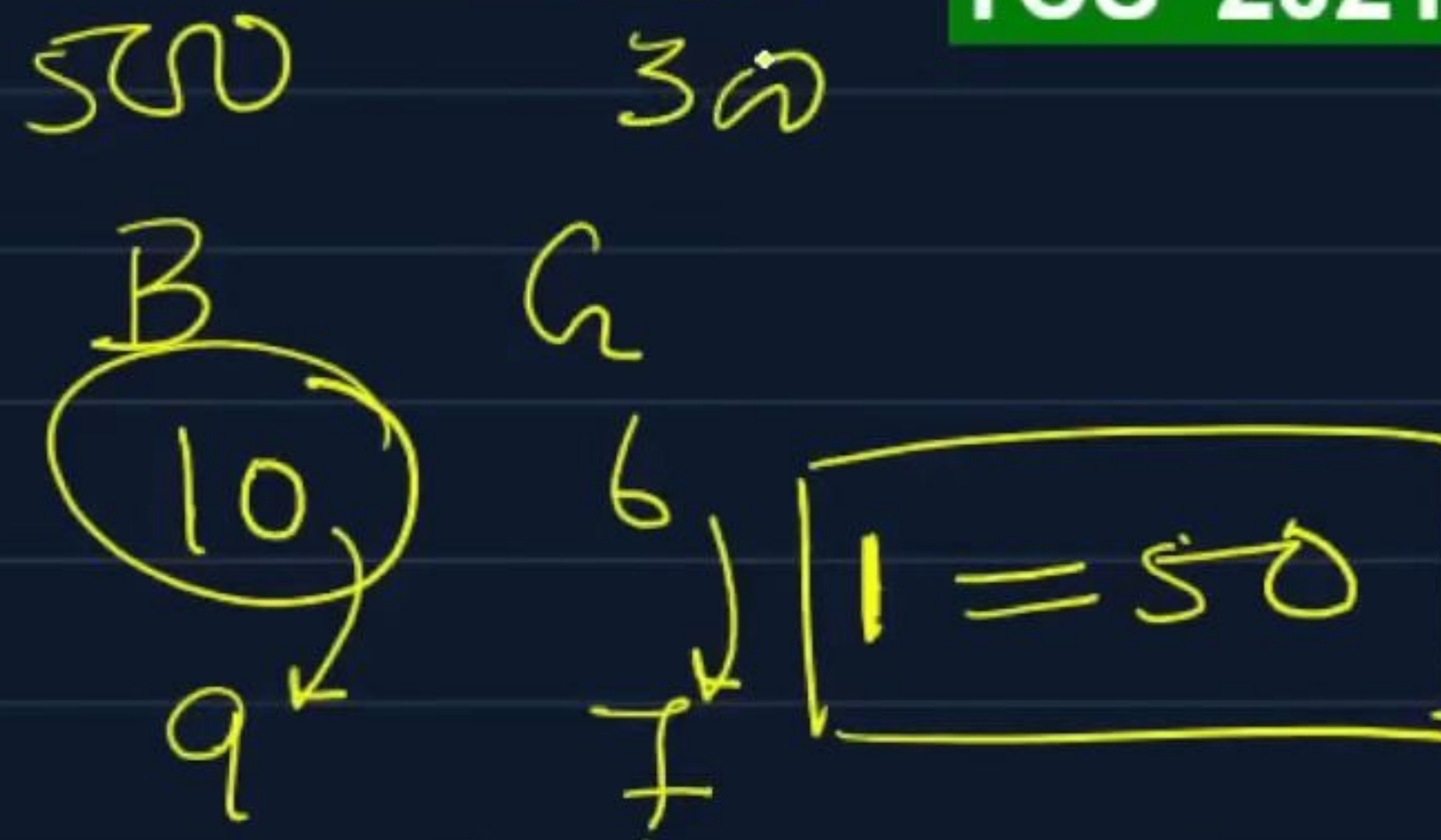
## Pre-Assessment :

The ratio of boys and girls in a college is 5 : 3. If 50 boys leave the college and 50 girls join the college, the ratio becomes 9 : 7.

The number of boys in the college is

- (1) 300
- (2) 400
- (3) 500 ✓
- (4) 600

TCS' 2021(2<sup>nd</sup> cycle)



00:50



## Question:

If  $C_1, C_2, C_3, C_4$ , and  $C_5$  are five companies.

The profits made by  $C_1, C_2$ , and  $C_3$  are in the ratio  $9 : 10 : 8$  while the profits made by  $C_2, C_4$ , and  $C_5$  are in the ratio  $18 : 19 : 20$ .

If  $C_5$  has made a profit of Rs 19 crore more than  $C_1$ , then the total profit (in Rs) made by all five companies is : **CAT 2017(1)slot** **TCS Digital type**

$$\begin{array}{ccccccccc} C_1 & C_2 & C_3 & C_4 & C_5 & & & & \\ \left( \begin{array}{c} 9 \\ 10 \\ 18 \end{array} \right) & & 8 & & & & & & \\ & & & (19) & 20 & & & & \\ \hline (81) & 90 & 72 & 95 & (100) & & & & \end{array}$$

A) 438 crore ✓

B) 435 crore

C) 348 crore

D) 345 crore

$$19 \Rightarrow 19 \text{ cr}$$

$$1 = 1 \text{ cr}$$

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The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

A. 3 : 3 : 10

TCS

B. 10 : 11 : 20

C. 23 : 33 : 60

D. Cannot be determined



$$\begin{array}{r}
 \textcircled{200} \quad 300 \quad 500 \\
 \textcircled{15\%} \quad | \quad 10\% \quad | \quad 20\% \\
 \hline
 230 : 330 : 600
 \end{array}$$

$$20 + 10$$



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$$\begin{array}{r}
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 \textcircled{15\%} \quad | \quad 10\% \quad | \quad 20\% \\
 \hline
 230 : 330 : 600
 \end{array}$$

20+10



The ratio of the number of boys to that girls in a village is 3:2. If 30% boys and 70% of girls appeared in an examination, the ratio of the number of students appeared in the examination to that not appeared in the same examination?

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Total

$$\begin{array}{r} B \\ 30. \\ | 30\% \\ \hline 9 \end{array}$$

$$\begin{array}{r} G \\ 20 \\ | 70\% \\ \hline 14 \end{array}$$

$$50$$

$$23$$

$$A. 23:27$$

$$B. 23:25$$

$$C. 21:25$$

$$D. 25:21$$

$$\frac{23}{27}$$

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**Question:.** A sum is divided among A, B, C and D such that the ratio of the shares of A and B is 2:3, That of B and C is 1:2 and that of C and D is 3:4. If the difference between the shares of A and D is ₹648, then the sum of the shares is:

- A) ₹2052      B) ₹2160      C) ₹2484      D) ₹1944

TCS' 2020(1<sup>st</sup> cycle)

How -





## Question

Rs. 11550 has to be divided between X, Y and Z such that X gets  $\frac{4}{5}$  of what Y gets and Y gets  $\frac{2}{3}$  of what Z gets. How much more does Z get over X (in Rs.)?

~~(1) 7200~~

~~(2) 1800~~

~~(3) 1170~~

**(4) 2450**

(SSC CGL Tier-II CBE (Exam.) 17.02.2018)

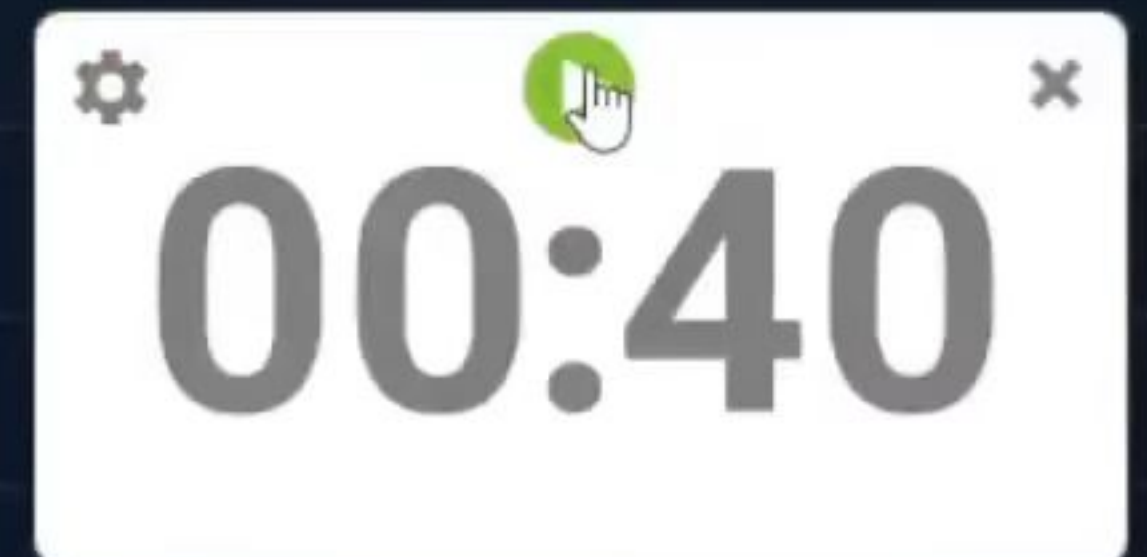
$$\frac{X}{Y} = \frac{4}{5} \quad \left\{ \quad \frac{Y}{Z} = \frac{2}{3} \right.$$

✓

|       |    |    |
|-------|----|----|
| X     | 4  | Z  |
| 4     | 5  | 5  |
| 2     |    | 3  |
| <hr/> |    |    |
|       | 10 | 15 |

7

7



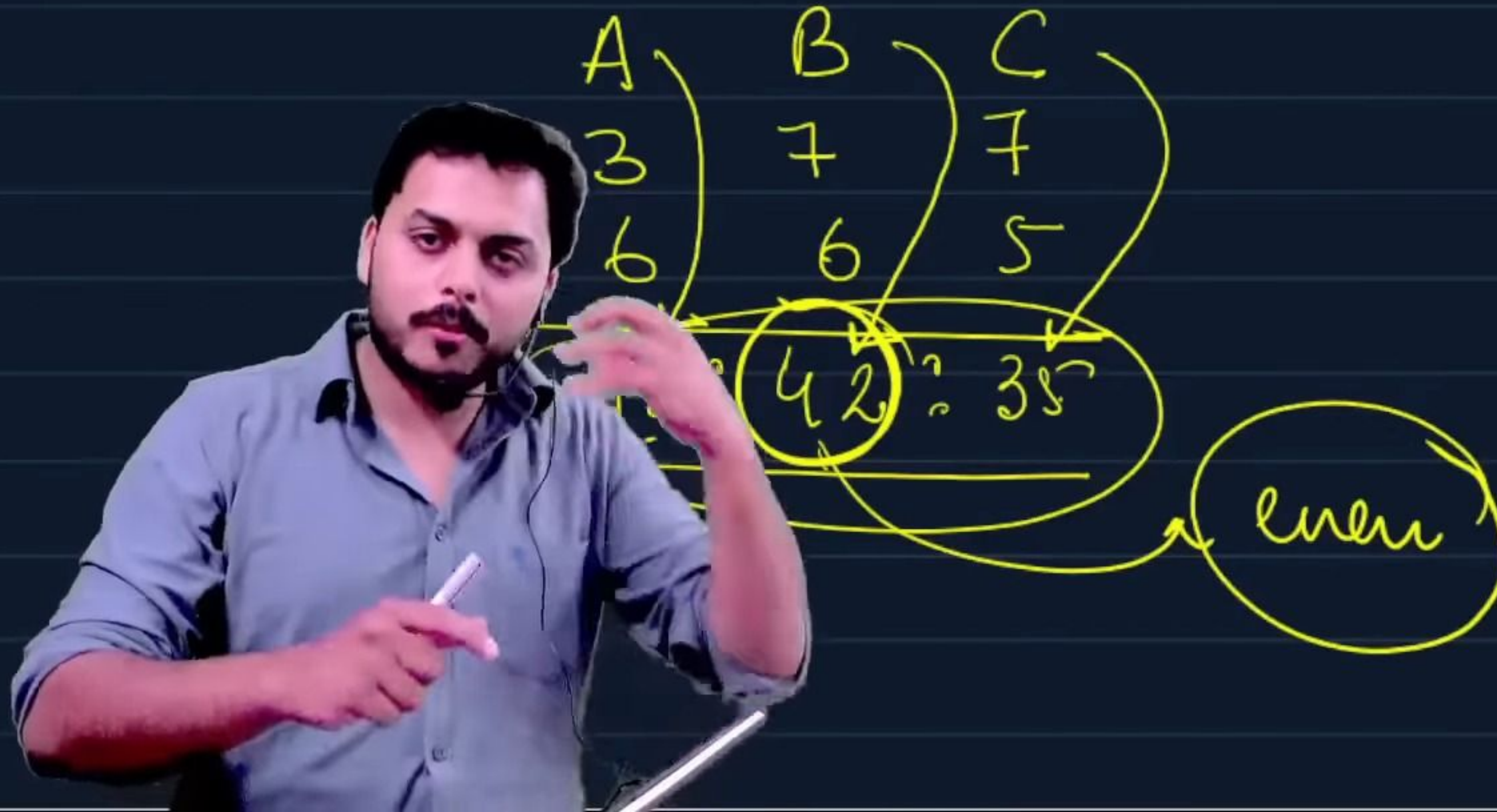


## Question

33,630 are divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3 : 7 and the ratio of the amount of B to that of C is 6 : 5 The amount of money received by B is

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- (1) 14,868 (2) 16,257 (3) 13,291 (4) 12,395



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An amount of Rs 735 was divided between A, B and C. If each of them had received Rs 25 less, their share would have been in the ratio 1:3:2.

The money received by C was ?

TCS NQT

- (a) Rs 195      (b) Rs 200      (c) Rs 220      (d) Rs 245

$$25 \times 3 = 75$$

|     |   |   |   |
|-----|---|---|---|
| 660 | 1 | 3 | 2 |
|-----|---|---|---|

$$\begin{array}{l} 6 \rightarrow 660 \\ 1 \rightarrow 110 \\ 2 \rightarrow \underline{220} + 25 \end{array}$$

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$$\begin{array}{c}
 S \\
 11
 \end{array}
 \quad
 \begin{array}{c}
 m \\
 7
 \end{array}$$

$$\begin{array}{l}
 4 = 4400 \\
 1 = 1100 \\
 7 = 7700
 \end{array}$$

A sum of money is divided between two person in the ratio  $2 : 9$  if the share of one person is 21 less than the other, find the sum ?

**Q.** Th price of a scooter and moped are in the ratio of 11:7. If a scooter costs Rs.4400 more than a moped, find the price of the moped?

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The ratio of two numbers is 3:8 and their difference is 115.

The smaller of the two numbers is :

1) 184

2) 194

3) 69

4) 59

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$$\begin{array}{cc} \text{I} & \text{II} \\ 3 & : 8 \end{array}$$

$$5 = 115$$

$$11 = 23$$

$$3 = 69$$



The speed of three cars in the ratio 2:3:4. the ratio between the times taken by these cars to travel the same distance is

$$S = \frac{D}{T} \rightarrow \text{Constant}$$

$$S \propto \frac{1}{T}$$

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Speed 2 : 3 : 4

Time  $\left( \frac{1}{2} \quad \frac{1}{3} \quad \frac{1}{4} \right) \times 12$

6 : 4 : 3 Ans.



| I | II | III |
|---|----|-----|
| 2 | 3  | 3   |
| 4 | 4  | 5   |
| 8 | 12 | 15  |



The sum of three numbers is 105. If the ratio of the first to second is 2 : 3 and that of the second to the third is 4 : 5, then the second number is?

36

$$\begin{array}{l} 35 \longrightarrow 105 \\ 1 \longrightarrow 105 \times 12 \\ \hline 35 \end{array}$$

\*k.  
H.W

The sum of three numbers is 98. If the ratio of the first to second is 2 : 3 and that of the second to the third is 5 : 8, then the second number

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$$a \Rightarrow 3$$

$$b+c \Rightarrow 9$$

$$c \Rightarrow 5$$

$$a+b \Rightarrow 7$$

$$4 \times 3$$

$$12 \times 1$$

$$a : b : c$$

$$3 : 4 : 5$$

$$\frac{4}{8} = \frac{1}{2}$$

## Question

If  $a : (b + c) = 1 : 3$  and  $c : (a + b) = 5 : 7$ , then  $b : (a + c)$  is equal to:

## Question

If  $a : (b + c) = 4 : 5$  and  $c : (a + b) = 7 : 11$ , then  $b : (a + c)$  is equal to

$$A : B$$

$$(2 : 3) \times 2$$

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## Question

If  $2A = 3B = 4C$ , then  $A : B : C$  is :

## Question

If  $2A = 3B = 5C$ , what is  $A : B : C$ ?

$$2A = 3B = 4C$$

$$\frac{A}{B} = \frac{3}{2} \quad \bigg| \quad \frac{B}{C} = \frac{4}{3}$$

$$\begin{array}{ccc} A & B & C \\ 3 & & \end{array}$$

$$\begin{array}{ccc} 2 & 4 & 3 \\ \hline 6 & 4 & 3 \end{array}$$

$$\frac{2A}{12} = \frac{3B}{12} = \frac{4C}{12}$$

$$\frac{A}{6} = \frac{B}{4} = \frac{C}{3}$$



## Question

If  $a : b = \frac{2}{9} : \frac{1}{3}$ ,  $b : c = \frac{2}{7} : \frac{5}{14}$  and  $d : c = \frac{7}{10} : \frac{3}{5}$  then  $a : b : c : d$  is

$$\frac{a}{b} = \frac{2}{9} \times \frac{3}{1} = \frac{2}{3}$$

$$\frac{b}{c} = \frac{2}{7} \times \frac{14}{5} = \frac{4}{5}$$

$$\frac{d}{c} = \frac{7}{10} \times \frac{5}{3} = \frac{7}{6}$$

|               | A     | B            | C            | D            |
|---------------|-------|--------------|--------------|--------------|
| $\frac{a}{b}$ | 2     | <del>3</del> | <del>3</del> | <del>3</del> |
| $\frac{b}{c}$ | 4     | 4            | 5            | 5            |
| $\frac{d}{c}$ | 2     | 6            | 6            | 7            |
|               | <hr/> |              |              |              |
|               | 16    | 24           | 30           | 35           |
|               | <hr/> |              |              |              |

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**Q.** If  $A : B = 2 : 3$  and  $B : C = 4 : 5$  and  $C : D = 6 : 7$ , then find the value of  $A : B : C : D$ ?

| A | B | C | D |
|---|---|---|---|
| 2 | 3 | 3 | 3 |
| 4 | 4 | 5 | 5 |
| 6 | 6 | 6 | 7 |

<sup>5 8</sup>  
A. ~~15:24:30:35~~

B. ~~16:24:30:35~~

C. ~~17:24:30:35~~

D. ~~18:24:30:35~~

E. ~~17:23:30:35~~

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## Question

If  $a : b = 7 : 9$  and  $b : c = 15 : 7$ , then what is  $a : b : c$

$$\begin{array}{ccc}
 A & B & C \\
 7 & \cancel{9} \times 3 & \cancel{9} \times 3 \\
 5 \times \cancel{15} & 15 & 7 \\
 \hline
 35 & : & 45 & : & 21
 \end{array}$$

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$$\frac{A}{B} = \frac{2}{3} \quad \frac{B}{C} = \frac{4}{5}$$

**Concept :** Group bazi :

|                                                                                                                                |   |     |   |     |
|--------------------------------------------------------------------------------------------------------------------------------|---|-----|---|-----|
| A                                                                                                                              |   | B   |   | C   |
| ↓ 2                                                                                                                            |   | 3 ↓ | → | 3 ↓ |
| ↓ 4                                                                                                                            | ← | 4 ↓ |   | 5 ↓ |
| <div style="display: flex; justify-content: space-around; width: 100%;"> <span>8</span> <span>12</span> <span>15</span> </div> |   |     |   |     |



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A : B : C  
5 : 11 : 13

6

An amount of Rs.950 is distributed among A, B, & C in the ratio of 5:11:3, what is the difference between the share of B and A.

(a) 300 (b) 340 (c) 500 (d) 700

#

The number of pupils of a is 55. The ratio of the numbers male pupils to the number of female pupils is

5 : 6. The numbers of female pupils is  
(1) 11 (2) 25 (3) 30 (4) 35

M : F  
5 : 6

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